

Jason Nguyen

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[LinkedIn Profile](#) | [Personal Website Portfolio](#)

CS graduate seeking a software or data engineering role, with research and internship experience spanning ML pipelines, distributed data systems, embedded systems, and full-stack development.

EDUCATION

SUNY University at Buffalo

August 2021 - May 2026

Bachelor of Science, Computer Science

Cicero-North Syracuse High School

September 2017 - June 2021

Graduated with Honors

EXPERIENCE

FedEx

January 2026 - May 2026

Trade Operations Analyst

- Built Excel automation workflows reducing manual customs processing time by 30%.
- Leveraged SQL, Excel reporting, and cross-platform systems to streamline brokerage processing and support time-sensitive customs clearance.

ARISE Adaptive Design

May 2024 - August 2024

Adaptive Design Intern

- Engineered and validated object-oriented firmware for 2 embedded systems (Arduino/C++) in a Linux environment, covering 23 unit test cases across sensor input, signal timing, and hardware-software integration.
- Reduced latency by 15% through iterative firmware optimization, using serial debug logs to isolate and resolve bottlenecks.
- Authored hardware configuration and debugging documentation that standardized setup across 3 device variants, cutting team onboarding and reconfiguration time.

Randstad

January 2024 - April 2024

Data Analyst Intern

- Cleaned and validated 9,000+ records using Excel (pivot tables, VLOOKUP, conditional logic) and SQL, resolving 4 categories of recurring data entry errors.
- Built 2 reporting templates that automated manual aggregation, reducing report turnaround by 1 hour per cycle.
- Conducted root cause analysis on data workflows and produced reports to reduce errors in weekly deliverables.

Booz Allen Hamilton

May 2023 - August 2023

Software Engineer Intern

- Developed internal web tools using React and Flask to streamline analyst workflows.
- Built REST API endpoints and SQL queries for internal dashboard and reporting systems.
- Delivered sprint-based features in an Agile environment using Git, JIRA, Docker, and CI/CD pipelines.

DataAnnotation

May 2022 - August 2022

Freelance Software Engineer

- Evaluated and refined AI-generated code across Python, JavaScript, and SQL to improve model accuracy.
- Designed structured prompts and test cases for LLMs, identifying edge cases and debugging incorrect outputs.
- Reviewed backend logic, API usage, and algorithmic solutions for technical correctness and engineering best practices.

United Radio

January 2020 - September 2020

Electronics Technician

- Diagnosed and repaired consumer electronics at 750 devices/week using circuit board-level fault analysis.
- Performed component-level soldering repairs guided by schematics, diagnostics, and repair manuals.
- Generated repair reports documenting fault findings, replaced components, and resolution outcomes across all processed devices.

SKILLS

ML/AI: PyTorch, TensorFlow, PySpark, OpenCV, NumPy, Pandas

Web/Full-Stack: React, Svelte, TypeScript, JavaScript, HTML/CSS, Docker, Node

Languages: Python, Rust, Java, C/C++, C#, Go, Scala, GDScript

Tools & Platforms: Git, Bash, MySQL, SQLite, PostgreSQL, MongoDB, Google Cloud, Arduino, Figma, JIRA

RESEARCH

Large-Scale Spotify Track Popularity Analysis with PySpark & ML | [Link to Report](#)

- Designed and executed a large-scale data analysis and machine learning pipeline on 97,624 Spotify tracks using PySpark on HDFS. Built a Random Forest Regressor to predict song popularity from audio features; improved R^2 by 41% (0.17 to 0.24) through hyperparameter tuning via 3-fold cross-validation. Performed genre popularity distribution analysis across 114 genres and audio feature significance testing (Welch's t-test, Cohen's d) across six popularity tiers, identifying acoustiness and danceability as the strongest predictors of viral performance.

Handwriting to Text: CRNN + LLM Study Guide Pipeline | [Link to Report](#)

- Built an end-to-end pipeline that converts handwritten student notes into structured study guides. Trained a Convolutional Recurrent Neural Network (CRNN) with CTC loss on 400k+ word images from the IAM Handwriting Database and Kaggle, achieving 12.47% Character Error Rate. Integrated Llama 3.3 70B via the Groq API for downstream summarization, reaching ROUGE-1 F1 of 0.423. Evaluated against Tesseract and BLIP-2 baselines.

PROJECTS

Personal Portfolio Website | *React, JavaScript, CSS, HTML, Git, Vercel* | <https://website-plum-six-56.vercel.app/>

- Built and deployed a personal portfolio website using React and Vercel, implementing a responsive single-page application with custom CSS styling and component-based architecture, serving as a centralized showcase of projects and professional background.

BDO Loot Tracker | *Python, Svelte, CSS, SQLite, Git* | <https://github.com/janhnguyen/BDO-Loot-Tracker>

- Built a desktop application using Python, Svelte, SQLite, Git, and packaged distribution tools with a modular architecture integrating UI, local data persistence, and data processing components to track and analyze in-game item data.

Lost Ark Logs | *Rust, Svelte, TypeScript, SQLite, Git* | <https://github.com/snoww/loa-logs>

- Designed and implemented UI/UX components for a Rust-based gameplay analytics tool, optimizing SQLite schema and queries to improve data storage and retrieval performance. Prioritized user feedback to develop product features. Owned feature development end-to-end within a multi-developer Git workflow, driving code reviews, issue tracking, and iterative releases.

Tic-Tac-Toe Bot | *Python, PyTorch* | <https://github.com/janhnguyen/rl446tictactoe>

- Developed a reinforcement learning framework in Python using PyTorch, implementing DQN, PPO, SAC, and Rainbow agents with a custom Tic-Tac-Toe environment, reward functions, and training pipeline to evaluate agent performance and learning efficiency.

OCR Party Inspection | *Python, PyTesseract, OpenCV, Numpy* | <https://github.com/janhnguyen/partyInspection>

- Built a Python application using OpenCV and PyTesseract to capture and process game client screenshots, extract structured data via OCR and image recognition, and transmit results to an external web service through automated data pipelines.

Social Media Site | *React, CSS, JavaScript, HTML, Git, Docker* | <https://github.com/janhnguyen/djibouti>

- Developed a full-stack social media web application using React and a custom backend API, implementing responsive UI components, RESTful client-server communication, and containerized deployment with Docker to ensure scalability, portability, and maintainable architecture.

LEADERSHIP & COMMUNITY INVOLVEMENT

Lost Ark Nexus Contributor | <https://www.lostark.nexus/> | <https://shorturl.at/18Kle>

- Authored and maintained a comprehensive technical guide analyzing system mechanics, optimization strategies, and performance tradeoffs for a large online gaming community.
- Interpreted gameplay metrics and iterative updates to refine recommendations. Actively supported users through Q&A and technical review.

UB Open Source Club

- Collaborated as an open source club member to develop indie games, contributing to coding, design, and testing, which enhanced project quality and innovation. Participated in brainstorming sessions and provided feedback.